



PRESIDENCY COLLEGE

(AUTONOMOUS)

AFFILIATED TO BENGALURU CITY UNIVERSITY, APPROVED BY AICTE, DELHI & RECOGNISED BY THE GOVT. OF KARNATAKA
RE-ACCREDITED BY NAAC WITH 'A+' GRADE

Student Register Number: 23P01114

Student Name: Sujay Mandal

Course: 3rd MCA

Subject: Software Engineering

Class & Section: MCA B

Certificate

This is to certify that Sujay Mandal has satisfactorily completed the course of assignment prescribed by the Presidency College (Autonomous) for the semester III MCA degree course in the year 2024 -2025.

MARKS	
MAX	OBTAINED

Signature of the Student
Member

Signature of the Staff

TRELLO AND CLEAR CASE TOOLS

Trello Tool

Definition:

Tasks, projects, and work teams may all be arranged and managed with Trello, an online project management platform. Trello, which was created by Fog Creek Software in September 2011, has grown to be one of the most widely used project management apps. Trello makes project management easier and more collaborative by using a system of boards, lists, and cards to help teams visualise their tasks and workflows.

Features:

Lists, cards, and boards:

- **Boards:** Stand in for specific tasks or procedures. There are several listings on each board.
- **Lists:** Show phases of a project or groups of related work. There are several cards in each list.
- **Cards:** Stand for specific assignments or things that must be finished. Cards may have labels, comments, due dates, checklists, attachments, and descriptions.

Comments and Attachments:

- Documents, images, and files can be attached to cards by users.
- Members of the team can leave comments on cards with questions, comments, or updates.

Dates of Due and Labels:

- Labels with colours aid in work classification and prioritisation.
- Setting deadlines helps guarantee that work is finished on schedule.

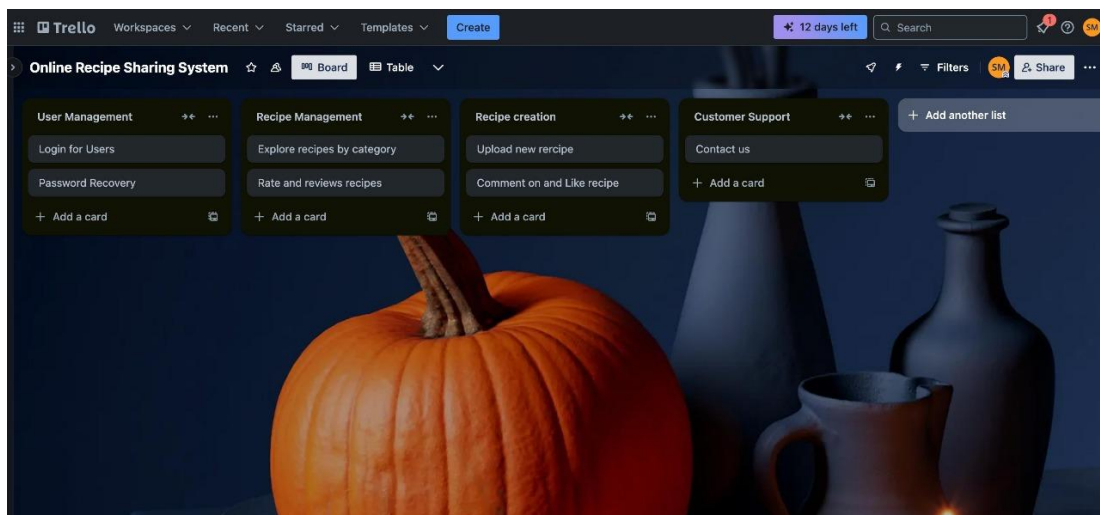
Using Butler for Automation:

- Butler's rules, buttons, and scheduled actions let users automate routine chores and workflows.

Integrations:

- Trello facilitates smooth collaboration and improved functionality by integrating with a large number of third-party platforms, including Slack, Google Drive, Dropbox, and more.

Screenshots of Trello Tool:



This screenshot shows the same Trello board in 'Table' view. The table lists the cards and their associated metadata, including the list name, labels, members, and due dates.

Card	List	Labels	Members	Due date
Login for Users	User Management	user id	SM	10 Mar
Password Recovery	User Management	forget password	SM	9 Mar
Explore recipes by category	Recipe Management	types of recipes	SM	31 Mar
Rate and reviews recipes	Recipe Management	rating on 1 to 5	SM	20 Mar
Upload new recipe	Recipe creation	name of recipe	SM	28 Mar
Comment on and Like recipe	Recipe creation	like and comment	SM	30 Mar
Contact us	Customer Support	support	SM	

Advantages:

- **Easy to Use:**
Users with different degrees of technical expertise can use Trello because of its user-friendly and intuitive visual interface.
- **Adaptable:**
Trello may be used to manage a wide range of projects, from intricate team projects to individual to-do lists.
- **Mobile-Friendly:**
Because Trello is accessible on several platforms—web, mobile, and desktop—users can access their projects from any location.
- **Cooperation:**
By offering real-time updates, notifications, and communication capabilities, Trello improves teamwork.

Disadvantages:

- **Limited Advanced Features:** Gantt charts, time tracking, and other sophisticated project management tools are not included.
- **Problems with scalability:** It might not be appropriate for very large organisations with intricate project requirements.
- **Over-Reliance on Integrations:** Third-party integrations are necessary for many advanced functionalities.
- **Limitations on Customisation:** There aren't many ways to alter the workflow and UI.
- **Security Issues:** Since this tool is cloud-based, some users may be concerned about data security and privacy.

Detailed analysis of the criteria while choosing Trello

Take into account the following factors while determining whether Trello is the best project management platform for your requirements:

Usability

- **Visual Task Management:** Trello's board, list, and card system can be a highly useful tool if you need one that provides a powerful visual representation of tasks and workflows.
- **Basic Project Management Needs:** Perfect for projects that are easy to moderately difficult and don't call for sophisticated tools like time tracking or Gantt charts.
- **Checklists and Attachments:** Helpful for attaching files straight to task cards and for breaking tasks down into smaller steps.

Ease to use:

- **User-Friendly Interface:** Team members of various skill levels may easily learn and utilise Trello thanks to its well-known simple and intuitive interface.
- **Minimal Training Needed:** Trello's ease of use might be a huge benefit if your team favours tools that need little to no training.

Cooperation

- Team members may easily communicate and work together directly within Trello thanks to its support for real-time updates, comments, and mentions.
- Trello makes it simple to share boards and manage permissions so that only authorised users can view and update tasks.

Power-Ups for Integration:

- Trello's Power-Ups allow for integration with a number of third-party apps, including Jira, Slack, Google Drive, and more, extending functionality to suit your unique requirements.
- **Automation:** To save time and automate repetitive processes, use Trello's Butler automation feature.

Scalability

- **Ideal for Small to Medium Teams:** Trello works effectively for projects and teams that are small to medium in size. For really complicated projects or very large organisations, it might not be the best option.

Cloud-Based Tool Security:

- Take into account how utilising a cloud-based tool may affect security. Trello offers enterprise-grade security, but it's crucial to determine whether it satisfies the data privacy and security requirements of your company.

Personalisation

- Flexible Workflow Management: Trello lets you modify boards, lists, and cards to suit your own project management approach and workflow.
- Limited Customisation Options: Trello may not give the degree of customisation required for highly specialised workflows, albeit offering limited customisation.

Community and Support

- Good Customer assistance: Trello provides a plethora of resources, such as tutorials, documentation, and an active user community, along with good customer assistance.
- Community Resources: Trello's vibrant user base and a wealth of online resources can assist with troubleshooting and maximising its functionality.

Clear Case Tool

Definition:

A software configuration management solution called IBM Rational ClearCase offers extensive version control, workspace management, and build auditing features. It is employed to oversee software development initiatives and guarantee uniformity among various configurations and versions.

Features:

Control of Versions

- File and Directory Versioning: Teams can monitor changes over time with ClearCase's ability to handle multiple versions of files and directories.

Baseline Management:

- Create baselines to record and save particular project iterations at significant turning points.

Branches and Combinations

- Branch Creation: Establish branches to support concurrent development initiatives. Developers can work on multiple features or bug fixes at once thanks to branching.
- Merge Operations: Ensure that all updates are seamlessly integrated by merging modifications from branches back into the mainline.

Perspectives

- Dynamic Views: Give developers access to the most recent versions of files and folders from the repository so they may work with the most recent data.
- Snapshot Views: For consistent development environments, make static copies of the project at predetermined intervals.

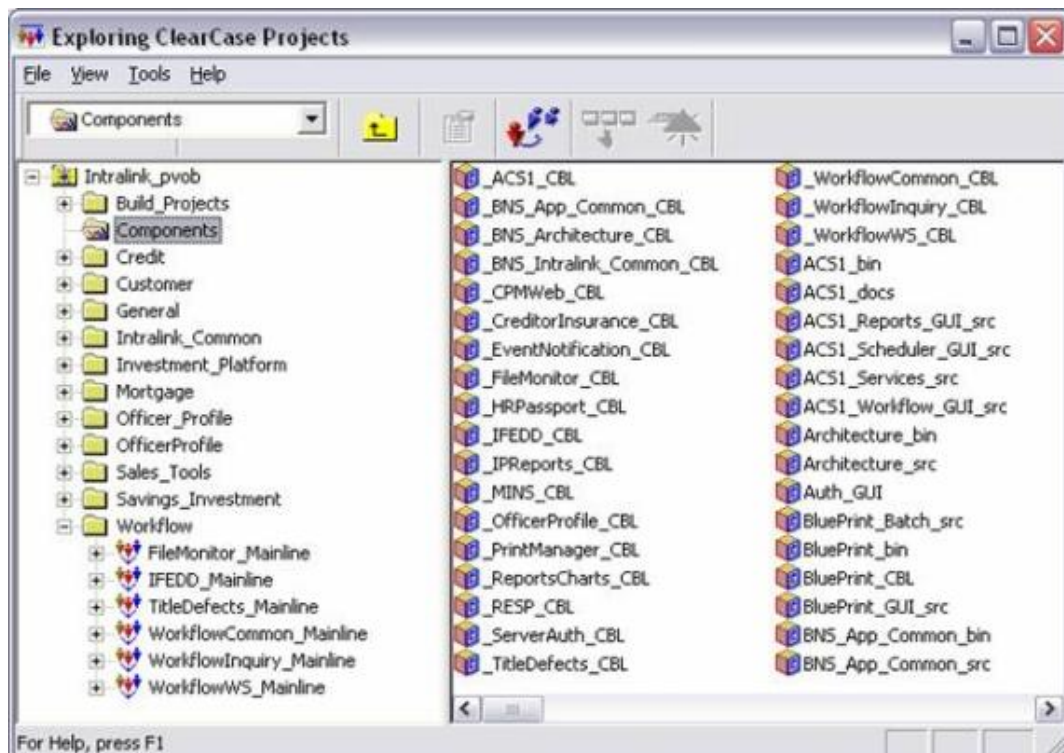
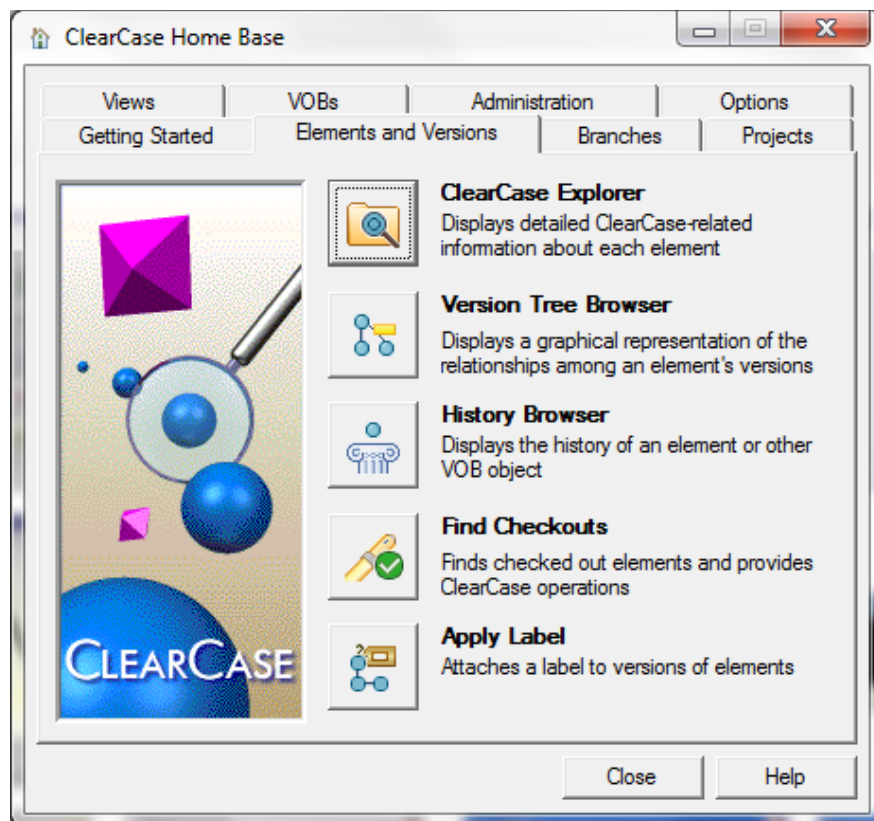
Labelling

- Version Labels: To facilitate identification and retrieval, provide labels to particular file versions. Teams can distinguish important versions, such as releases or milestones, with the use of labels.

Including

- Integration of Development Tools: Works with a variety of development environments and tools, such as IDEs and CI/CD pipelines.
- Build Integration: Facilitates automated build and deployment procedures by supporting integration with build systems.

Screenshots:



Advantages:

- **Complete Version Control:** Provides strong version control features to handle modifications over time.
- **Supports intricate branching and merging scenarios** for concurrent development projects.
- **Workspace Management:** Offers advanced workspace management tools to manage various setups.
- **Build auditing:** Monitors and examines builds to guarantee quality control and compliance.
- **Integration:** Works nicely with a variety of development environments and other IBM Rational technologies.

Disadvantages:

- **Complexity:** Needs a lot of setup and upkeep and has a steep learning curve.
- **Performance problems:** Particularly for big projects, it can be resource-intensive and slow.
- **Cost:** High maintenance and license costs.
- **Limited Offline Functionality:** Most actions require a network connection, which restricts offline functionality.
- **Compared to distributed version control systems (DVCS) like Git, centralised VCSs are less adaptable.**

Detailed analysis of the criteria while choosing Clear Case

Take into account the following factors when determining whether IBM Rational ClearCase is the best configuration management solution for your requirements:

Functionality Version Control: ClearCase is a good choice if you require strong version control features to handle many iterations of files, directories, and complete projects.

- **Branching and Merging:** Perfect for intricate branching and merging situations, this feature makes it possible to effectively coordinate concurrent development initiatives.

- **Workspace Management:** Facilitates various configurations and separate development environments by supporting advanced workspace management features.
- **Build Auditing:** ClearCase offers extensive monitoring and auditing features if you need thorough build auditing to guarantee compliance and quality assurance.
- **Integration:** Works with IDEs, CI/CD pipelines, and other IBM Rational products, among other development environments and tools.

Usability

- **Complexity:** Expect a challenging setup procedure and a steep learning curve. Effective usage of ClearCase necessitates extensive training and experience.
- **User Training:** Make sure your group has the time and resources necessary to devote to becoming proficient using ClearCase.

Scalability

- **Big Projects:** Fit for big businesses and projects that need sophisticated configuration management and version control.
- **Enterprise-Level needs:** ClearCase is capable of managing the complexity and scope of enterprise-level configuration management needs.

Integration Development Environment Integration:

- **Verify that it works with the IDEs, CI/CD pipelines, and development tools you currently use.** ClearCase works well with a variety of development environments and other IBM Rational technologies.
- **Build System Integration:** To enable automated build and deployment procedures, make sure ClearCase can interface with your build systems.

Cost of Maintenance and Licensing:

- **Take maintenance and licensing costs into account.** Because ClearCase can be costly, make sure the advantages match your project's needs and budget.

Enterprise-Grade Security:

- Determine whether ClearCase satisfies the data privacy and security requirements of your company. ClearCase offers robust security measures that are appropriate for business use.

Community and Support

- Full Support: IBM provides ClearCase users with a wealth of training materials and support. Take into account the availability of training materials and customer service to help with usage and deployment.
- Community Resources: Although not as popular as some other programs, there are still online resources for best practices and troubleshooting as well as a user community.

Personalisation

- Advanced Customisation Options: Determine whether ClearCase offers the degree of personalisation required for your particular procedures and workflows. ClearCase provides sophisticated customisation choices to meet specific needs.

Criteria for choosing the Right Tool

Take into account the following factors when deciding between Trello and ClearCase:

Functionality: Verify that the program offers the functions you require for project management.

Ease of Use: Take into account the tool's ease of use and learning curve.

Scalability: Assess the tool's ability to grow with your company and manage its growing complexity.

Integration: Verify that the tools and systems you use are compatible and integrated.

Cost: Take into account the pricing structure and make sure you can afford it.

Security: Evaluate the tool's security and data privacy features.

Support: To help with implementation and usage, look for quality customer service and training materials.

Customisation: Verify that the tool can be tailored to your unique procedures and workflows.

Criteria	Trello	Clear Case
Functionality	Visual organisation and task management	Controlling versions, branching, and merging
Usability	Easy to use and intuitive	High and complicated learning curve
Scalability	restricted for big businesses	Ideal for big projects
Integration	Connects to a number of external tools	Complements IBM Rational Tools
Cost	Cost-effective, freemium business model	High maintenance and licensing costs
Security	Cloud-based security and possible data issues	Robust security and enterprise-level
Support	Excellent support and local resources	Complete assistance from IBM
Customization	Minimal personalisation	More choices for customisation